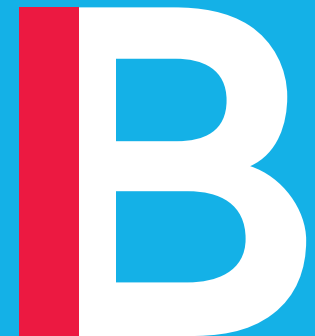


Quantitative Methods

Review

by Professor Shuang Zhang





Remember our objectives?

- Rethink about cause-and-effect questions.
- Approach *your* cause-and-effect questions coolly, however, not armed with passion but with data.
- Equip you with *disciplined* data analysis methods used in economics and business.

The key in the path from cause to effect:

Ceteris Paribus: Other Things Equal

Is it possible?

Potential outcomes

We are interested in the relationship between “treatment” and some outcome that may be impacted by the treatment

For each individual, there are two potential outcomes:

- $Y_{1,i}$ = i's outcome if she receives treatment
- $Y_{0,i}$ = i's outcome if she **had not received** treatment

The fundamental problem of causal inference:

We never observe both potential outcomes for the same individual

Average Causal Effects

What we actually want to know is the **average causal effect**, but that is not what we get from a difference in means comparison

Difference in group means

= average causal effect of program on participants + selection bias

Selection Bias

The goal of most empirical economic research is to overcome selection bias, and therefore to say something about the causal effect of a variable like D_i . (Angrist & Pischke, 2009, p.15)

Method #1: Randomized experiments

- Treatment is randomly assigned
- **Assumption**: treatment group and control group are similar in the absence of the treatment
- When the assumption is valid, selection bias is zero
 - Difference in group means = average causal effect
- Internal validity vs. External validity
 - Internal validity: the causal effect is valid within the experimental sample
 - External validity: does the same effect apply in another sample/context?

Method #2: Basic regression

- A linear regression model:

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

- **Assumption**: Conditional Independence Assumption (CIA)
 - Conditional on observable characteristics X , potential outcomes are independent of treatment status D
- We include characteristics X into the regression model:
$$Y_i = \alpha + \beta D_i + \gamma X_i + \varepsilon_i$$
 - If CIA holds, conditional on X , treatment D is uncorrelated with error term (as good as randomly assigned)
 - β is causal

Correlation does not prove causation

What does prove causation?

- Gold standard: randomized experiment
- Least suggestive: basic regression
- In between: better non-experimental paths to *Other Things Equal*
 - Instrumental Variable
 - Regression Discontinuity Design
 - Difference-in-Difference

Method #3: Instrument variable (IV)

IV runs two regressions separately:

$$Y_i = \alpha + \rho Z_i + \varepsilon_i \text{ (Reduced form)}$$

$$D_i = \alpha + \phi Z_i + \mu_i \text{ (First stage)}$$

$$\lambda_{IV} = \rho/\phi$$

- **Assumptions**

- There is a first stage: the IV affects the causal channel of interest
- Independence assumption: IV is as good as randomly assigned, or to say IV is uncorrelated with the error term

Method #4: Regression Discontinuity (RD)

RD uses a regression like

$$Y_a = \alpha + \beta D_a + \gamma a + \varepsilon_a$$

- Running variable: a
 - Treatment switches sharply on or off as the running variable passes a cutoff
- **Assumption**: D_a is solely determined by a
 - By controlling for a , NO omitted variable correlated with D_a in the error term
- When is RD invalid?
 - Manipulation of the running variable
 - Imperfect enforcement of the rule

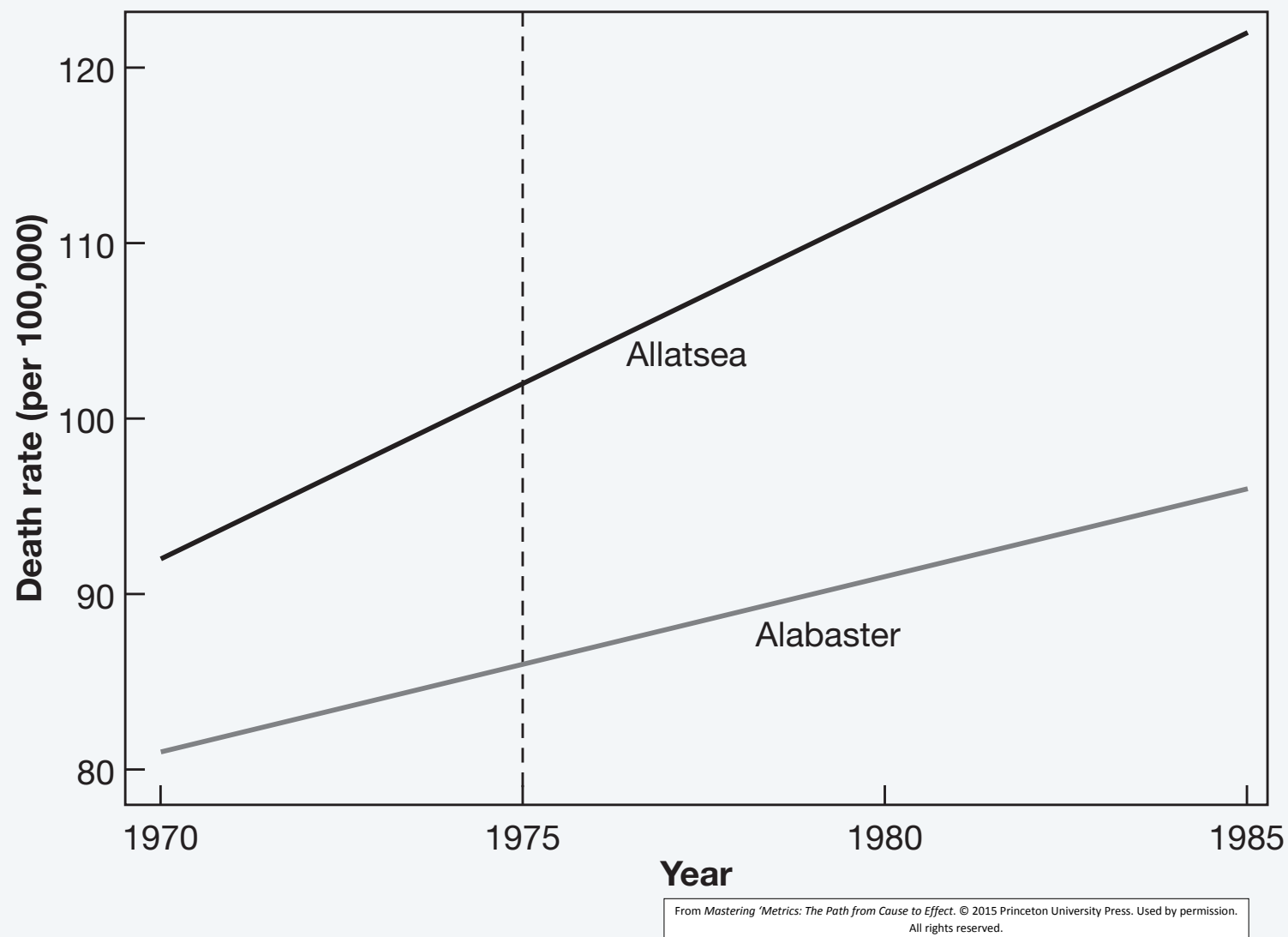
Method #5: Difference-in-Difference

	Treatment Group With Program	Control Group Without Program	Difference Across Groups
After Intervention	Average of outcome Y_A^T	Average of outcome Y_A^C	Single-Difference $Y_A^T - Y_A^C$
Before Intervention	Average of outcome Y_B^T	Average of outcome Y_B^C	$Y_B^T - Y_B^C$
Difference across time	$Y_A^T - Y_B^T$	$Y_A^C - Y_B^C$	Double-Difference $(Y_A^T - Y_A^C) - (Y_B^T - Y_B^C)$

Difference-in-Difference: assumption

- It is OK that treatment group and control group don't have the same **level** of outcome before the treatment
- **Assumptions:**
 - Common trends: absent any treatment, the trend in the treatment group is what we should have expected to see in the control group
 - No other relevant policies or changes around the same time

FIGURE 5.5
A spurious MLDA effect in states where trends are not parallel



Mission impossible, but there is hope

Always be critical of the underlying assumptions of any method!

Module Excellence Survey (MODES)



We are going to ask you to spend the next 10 minutes providing feedback on your module. Your feedback is very helpful in enhancing student experiences here at Imperial.

The instructor will leave the room whilst you provide feedback and your feedback is anonymous and instructors will only see your feedback once grading of assessment is complete.

- Please check your email now for the survey link (hosting@evasys.co.uk) or use the QR code above.
- There is a survey for each instructor who taught more than 3 hours on the module (including Teaching Assistants, where applicable)
- Please:
 - Ensure your feedback is constructive
 - Identify areas of good practice, as well as any suggestions for improvement.
 - Note that the scale runs from negative on the left (strongly disagree) to positive on the right (strongly agree)
- Your feedback will be passed to the instructors, Programme Team and the School's senior management for consideration.
- Programme Teams will provide a summary response to the feedback received from your cohort.

Thank you.